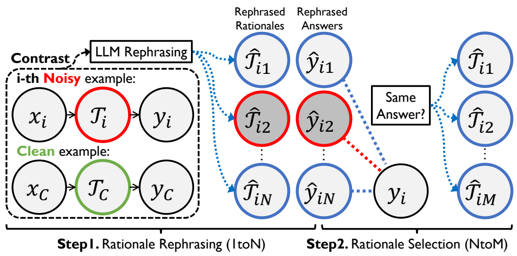

Figure 4: CD-CoT's first two steps for data denoising. First, it rephrases the i-th noisy example by contrasting it with the clean example. Then, with the obtained N rephrased examples, it selects the M qualified candidates by checking the validity of the rephrased answers $\hat{y}_{i1}, \ldots, \hat{y}_{iN}$ *w.r.t.* y_i.

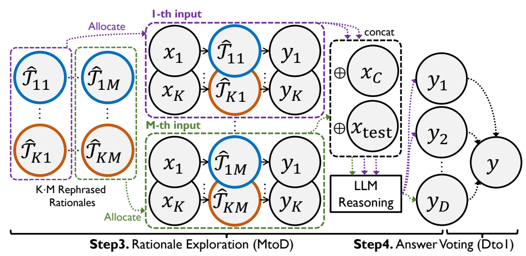

Figure 5: CD-CoT constructs M inputs (K-shot) by allocating the $K \cdot M$ rephrased rationales. These inputs are concatenated with the clean example and test question and then fed to an LLM for reasoning separately. The obtained D answers are equally voted to obtain the final answer y.

Observation 4.5. **Different LLMs are generally vulnerable to noisy rationales.** In Tab. 6, we evaluate different LLMs across three settings: 0-shot CoT, 3-shot clean rationales, and 3-shot medium-level noisy rationales. Notably, Gemini-Pro outperforms GPT-3.5 in overall performance. However, it demonstrates a similar degree of sensitivity to noise, with a 2.4%-15.7% performance decline with irrelevant rationales and a 7.8%-66.8% decline with inaccurate rationales compared to clean rationales. While Mixtral 8x7B shows a slight underperformance compared to GPT-3.5, it also manifests a vulnerability to noise, incurring a 1.4%-11.2% loss with irrelevant rationales and a greater 4.2%-23.8% loss with inaccurate rationales. By contrast, Llama2-70B performs suboptimally, with a 0.4%-2.0% drop for irrelevant thoughts and a larger 1.0%-2.2% drop for inaccurate thoughts.

Further investigation. Inspired by Min et al. [54], we further explore the mapping among questions, rationales, and answers through shuffling experiments. Specifically, given the 3-shot prompting examples $\{(x_1, \mathcal{T}_1, y_1), (x_2, \mathcal{T}_2, y_2), (x_3, \mathcal{T}_3, y_3)\}$, we test three configurations, *i.e.*, shuffle questions $\{(x_1, \mathcal{T}_3, y_3), (x_2, \mathcal{T}_1, y_1), (x_3, \mathcal{T}_2, y_2)\}$, shuffle answers $\{(x_1, \mathcal{T}_1, y_3), (x_2, \mathcal{T}_2, y_1), (x_3, \mathcal{T}_3, y_2)\}$, and shuffle rationales $\{(x_1, \mathcal{T}_3, y_1), (x_2, \mathcal{T}_1, y_2), (x_3, \mathcal{T}_2, y_3)\}$. These break the original mappings. The results under these configurations are shown in Tab. 7, which induces the following observation.

Observation 4.6. **Shuffling the mappings of prompting examples degenerates the reasoning but still performs better than without prompting.** This means that while LLMs may not heavily rely on the exact mapping (of question, rationale, and answer), they still benefit from demonstrating information even with shuffling. Notably, this finding is consistent with the conclusions of [54] that LLMs learn more abstract task information from the demonstrations rather than simply memorizing question-answer pairs. *More importantly, LLMs are less vulnerable to shuffled mappings than noisy rationales*. Unlike shuffling, the irrelevant or inaccurate information in noisy rationales introduces misleading elements that significantly interfere with the model's ability to learn *correct* task patterns, thereby resulting in more severe performance degradation. This extends [54]'s finding and shows that the quality of reasoning steps can be more crucial than the exact mapping of prompting examples.

5 Method

This section aims to enable LLMs to discern and remove noisy thoughts. The observations in Sec. 4 and previous works show that current LLMs cannot achieve this with their intrinsic denoising ability, even enhanced with self-denoising methods. Therefore, we would claim that the *external supervision* is necessary for enhancement, which should be sufficient for denoising and accessible in practice. Existing methods with external supervision [29, 81, 9] require (1) oracle feedback on the test question, (2) human feedback of errors on specific tokens or positions, or (3) expert knowledge to construct detailed descriptions of specific tasks. By contrast, we believe that *a clean CoT demonstration* is more attainable and practical, which can be the *minimal requirement* for denoising-purpose prompting.

Therein, we assume that LLMs *can* identify noisy thoughts by *contrasting* a pair of noisy and clean rationales and discerning their differences, similar to contrastive learning [26, 6, 35]. Here, the denoising power could come from the abilities of the instruction following and step-by-step reasoning [84, 66]. Hence, we propose the framework of CD-CoT, **C**ontrastive **D**enoising with noisy **CoT**. The design principle is to explore and then exploit, *i.e.*, (1) rephrasing and selecting rationales in input space to achieve explicit denoising, and then (2) exploring diverse rationales and voting answers in output space for deriving the final answer, as in Figs. 4 & 5. The details are as follows.

Algorithm 1 CD-CoT: Contrastive Denoising with Noisy Chain-of-Thought.

Require: an LLM f_θ, the prompt of contrastive denoising $\mathcal{P}_{\text{denoise}}$, one test question x_{test}, one clean example $(x_{\text{C}}, \mathcal{T}_{\text{C}}, y_{\text{C}})$, K prompting examples $S_n = \{(x_i, \mathcal{T}_i, y_i)\}_{i=1}^K$, hyper-parameters N, M, and reasoning budget $\{B_i\}_{i=1}^M$ (satisfies that $\Sigma_{i=1}^M B_i = D$, where D is the total budget).

for $i = 1 \ldots K$ **do**
  initialize the set of rephrased results of i-th example $\mathcal{R}_i \leftarrow \emptyset$.
  for $j = 1 \ldots N$ **do**
    `# Step-1: Rationale Rephrasing via Supervised Contrasting`
    obtain a rephrased example as $(x_i, \hat{\mathcal{T}}_i, \hat{y}_i) \leftarrow f_\theta\Big(\mathcal{P}_{\text{denoise}}(x_{\text{C}}, \mathcal{T}_{\text{C}}, y_{\text{C}}, x_i, \mathcal{T}_i, y_i)\Big)$.
    if match answer $\hat{y}_i = y_i$, then store the rephrased example as $\mathcal{R}_i \leftarrow \mathcal{R}_i \cup \{(x_i, \hat{\mathcal{T}}_i, \hat{y}_i)\}$.
  end for
  `# Step-2: Rationale Selection`
  randomly select M rephrased examples from $\mathcal{R}_i$ and obtain $\tilde{\mathcal{R}}_i = \{(x_{is}, \hat{\mathcal{T}}_{is}, \hat{y}_{is})\}_{s=1}^M$.
end for
`# Step-3: Rationale Exploration`
initialize the set of answers $\mathcal{Y} \leftarrow \emptyset$.
for $i = 1 \ldots M$ **do**
  construct an input $\mathcal{P}_i \leftarrow \{(x_{ji}, \hat{\mathcal{T}}_{ji}, \hat{y}_{ji})\}_{j=1}^K$, where $(x_{ji}, \hat{\mathcal{T}}_{ji}, \hat{y}_{ji})$ is the i-th element of $\hat{\mathcal{R}}_j$.
  concatenate $\mathcal{P}_i$ with the clean example and test question as $\mathcal{P}_i \leftarrow \mathcal{P}_i \cup \{(x_{\text{C}}, \mathcal{T}_{\text{C}}, y_{\text{C}}), x_{\text{test}}\}$.
  for $j = 1 \ldots B_M$ **do**
    get one answer by LLM reasoning as $y_j \leftarrow f_\theta(\mathcal{P}_i)$.
    store the answer as $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{y_j\}$.
  end for
end for
`# Step-4: Answer Voting`
initialize the dictionary of answer count $\mathcal{C}$ that $\forall y_j \in \mathcal{Y}, \mathcal{C}[y_j] = 0$.
for $j = 1 \ldots D$ **do**
  update $\mathcal{C}[y_j] \leftarrow (\mathcal{C}[y_j] + 1)$.
end for
get the final answer y with maximum counts as $y \leftarrow \arg\max_y \mathcal{C}[y]$.
return the answer y.

5.1 Implementation

Step-1: Rephrasing via Supervised Contrasting (1 to N). First, we establish a general prompt of contrastive rephrasing to construct a pair of contrastive examples, as shown in the template below. This steers the model towards learning from the clean example and then rephrasing and rectifying the noisy examples. To be specific, given one clean example and K noisy examples, we generate N rephrased rationales for each noisy example independently and obtain $K \cdot N$ rephrased rationales.

> **Prompt of Contrastive Rationale Rephrasing:**
> Here are two examples for the same type of task: the first example has correct explanation and correct answer, and the second example has distracted explanation and correct answer. Please follow the first example and give me the correct explanation and answer for the second example, which should be logically consistent with the first one.
> **First Example:** Q: [Question], E: [Explanation], A: [Answer].
> **Second Example:** Q: [Question], E: [Explanation], A: [Answer].

Step-2: Rationale Selection (N to M, $N \geq M$). Next, we employ answer matching to select those rephrased examples with unchanged answers, leaving behind a refined candidate pool. Subsequently, we randomly select M rephrased rationales from the pool and concatenate them to form the contexts.

Step-3: Rationale Exploration (M to D, $M \leq D$). For the M different contexts, we explore rationales by repeated reasoning with the budget of D reasoning repetitions. Notably, a higher temperature parameter, *e.g.*, 1, is set to introduce more randomness in generating diverse rationales.

Step-4: Answer Voting (D to 1). Ultimately, all the D answers are equally voted into a final answer.

Instantiation. By tuning the hyper-parameters N, M, and D, we balance exploration and exploitation in the input and output space. The overall procedure of our proposed CD-CoT is presented in Algorithm 1. Besides, we further explain the details of each step of this algorithm in Appendix E.2.

Task	Method $\mathcal{M}$	Additional Information	$\mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{clean}})$	$\mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{irrelevant}})$				$\mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{inaccurate}})$			
				Easy	Medium	Hard	Avg.	Easy	Medium	Hard	Avg.
Math Base-9	Base	-	46.4	39.3	30.3	26.6	32.1	23.2	10.1	6.0	13.1
	w/ SCO [29]	Ground Truth	53.6	46.3	39.6	36.4	40.8	34.7	22.0	17.7	24.8
	w/ BT [81]	Noise Position	47.2	39.2	34.2	29.9	34.4	30.1	18.4	14.1	20.9
	w/ CC [9]	Clean Demo	44.9	43.3	44.6	45.5	44.5	37.2	31.7	30.7	33.2
	w/ CD-CoT (ours)	Clean Demo	**60.7**	**59.7**	**60.7**	**57.2**	**59.2**	**54.0**	**58.7**	**48.4**	**53.7**
Math Base-11	Base	-	23.9	19.1	13.6	10.7	14.5	14.0	6.7	3.6	8.1
	w/ SCO [29]	Ground Truth	**33.0**	29.2	24.0	20.0	24.4	**29.2**	20.0	17.2	22.1
	w/ BT [81]	Noise Position	24.3	17.9	17.2	13.7	16.3	12.8	9.2	6.8	9.6
	w/ CC [9]	Clean Demo	22.3	19.1	18.4	18.2	18.6	19.0	15.3	14.6	16.3
	w/ CD-CoT (ours)	Clean Demo	31.0	**33.7**	**32.7**	**34.7**	**33.7**	29.0	**30.7**	**25.3**	**28.3**
Symbolic Equal	Base	-	32.7	28.1	25.1	23.0	25.4	29.1	26.1	22.7	26.0
	w/ SCO [29]	Ground Truth	38.5	34.9	33.4	32.7	33.7	34.0	34.1	34.5	34.2
	w/ BT [81]	Noise Position	31.8	26.0	22.7	22.6	23.8	26.3	22.7	22.9	24.0
	w/ CC [9]	Clean Demo	37.8	33.8	32.7	32.0	32.8	31.3	33.0	29.9	31.4
	w/ CD-CoT (ours)	Clean Demo	**42.7**	**44.7**	**42.7**	**44.0**	**43.8**	**42.6**	**41.3**	**42.7**	**42.2**
Symbolic Longer	Base	-	9.2	6.3	7.2	6.0	6.5	7.0	6.8	6.0	6.6
	w/ SCO [29]	Ground Truth	**18.7**	**12.1**	10.5	11.3	11.3	**15.2**	**15.9**	9.8	**13.6**
	w/ BT [81]	Noise Position	7.2	3.4	3.5	2.5	3.1	3.8	3.6	3.6	3.7
	w/ CC [9]	Clean Demo	9.4	9.8	7.9	7.9	8.5	8.5	7.4	6.5	7.5
	w/ CD-CoT (ours)	Clean Demo	12.3	12.0	**12.0**	**13.0**	**12.3**	12.3	10.0	**11.0**	11.1
Commonsense	Base	-	45.7	44.3	42.3	41.4	42.7	36.7	33.4	28.3	32.8
	w/ SCO [29]	Ground Truth	**63.5**	**60.1**	**56.1**	**60.3**	**58.8**	**56.2**	**58.5**	**57.9**	**57.5**
	w/ BT [81]	Noise Position	47.7	23.5	28.3	32.5	28.1	11.6	11.0	15.8	12.8
	w/ CC [9]	Clean Demo	48.3	45.7	43.6	44.0	44.4	42.1	40.8	40.5	41.1
	w/ CD-CoT (ours)	Clean Demo	49.0	50.3	54.7	50.3	51.8	51.0	49.7	49.7	50.1

Table 8: Performance of denoising methods that require additional information for supervision.

Theoretical analysis. To understand the underlying mechanism of CD-CoT, we also conduct the theoretical analysis based on the distinguishability [90] of in-context learning. The full analysis is in Appendix D, where we find that the noisy demonstration in ICL can decrease the distinguishability of in-context matching with the clean-prompt distribution, while our method can mitigate this issue. Besides, we build a self-supervised variant of CD-CoT and empirically evaluate it in Appendix F.7.

5.2 Empirical Study

In this part, we empirically verify the effectiveness of CD-CoT and start by introducing the baselines.

Baseline methods. We employ three methods that require *additional information*: (1) Self-Correction with Oracle Feedback (SCO) [29] utilizes the *ground truth answers* of test questions to determine when to terminate the self-correction loop; (2) Backtracking (BT) [81] guides self-correction by providing the model with the *position* of the first noisy thought; (3) Contrastive Chain-of-Thought (CC) [9] conducts direct reasoning with all the noisy or *clean examples* without implicit or explicit denoising.

Main results. As in Tab. 8, CD-CoT demonstrates a significant performance improvement across all datasets, with an average improvement of 17.8% compared with the base model under noisy settings. Notably, on Math-Base-9, Math-Base-11, and Symbolic-Equal, CD-CoT surpasses all baseline methods by a significant margin. On Symbolic-Longer and Commonsense, CD-CoT only slightly lags behind SCO. However, SCO requires the ground truth answer to the test question, which should be unknown in practice, as pointed out in [29]. In comparison, CD-CoT only necessitates an additional clean demonstration, making it much more practical to apply across realistic scenarios. *Notably, CD-CoT outperforms SCO in* 20 *out of* 30 *settings and surpasses BT, CC in all* 30 *settings*.

Besides, CD-CoT displays *remarkable resistance* to the magnitude of noise. Therein, CD-CoT demonstrates enhanced resilience against inaccurate noise on mathematical tasks, which are quite challenging. For instance, on Math Base-9 with inaccurate rationales, the average accuracies of SCO and BT decline significantly by 28.8% and 26.3% compared to the accuracies with clean rationales. In contrast, CD-CoT exhibits a more modest decline of 7.0%. An ablation study of components in Appendix F.3 demonstrates the denoising power and performance gain of CD-CoT, attributed to its contrastive denoising with rationale rephrasing as well as repeated reasoning with voting components.

Ablation study of varying hyper-parameters. By manipulating the values of N, M, D, and C, we generate diverse algorithm instances. Here, D denotes the reasoning times allocated to the M inputs, while C signifies whether the clean example is used in step 3. As demonstrated in Tab. 9, the clean example utilized by CD-CoT during the reasoning process plays a pivotal role. The omission of this clean example results in an average decrease of 3.3% and 4.5% in accuracy under irrelevant noise and inaccurate noise, respectively. Besides, the accuracy exhibits subtle variations when employing different algorithm instances, with the highest average accuracy observed at 51.3% and the lowest average accuracy at 49.3%. Further, Tab. 10 presents the average number of tokens used in reasoning. We set $M = 2$ to strike a balance. Please refer to Appendix E.3 for detailed hyper-parameter selection.